

Abstract No.

Better Than Hiding: Gliomas Reprogram the Immune System to Disarm Host Defense

irene appolloni^{1,2,}, Francesca Piaggio², Giorgia Guccione², Davide Ceresa^{1,2}, Elena Causa^{1,2}, Sebastian Garcia Mena^{1,2}, Daniela Marubbi^{1,2}, Fabrizio Loiacono², Chiara Riviera², and Paolo Malatesta^{1,2}*

¹University of Genova

²AOM-IRCCS Ospedale policlinico San Martino

*Presenting author

Glioma progression is accompanied by profound changes in the interaction between tumor cells and the immune system. However, the mechanisms through which high-grade gliomas escape immune control remain incompletely understood. Here, we investigated this process using a PDGF-B-driven murine model in which low- and high-grade stages of gliomagenesis can be experimentally distinguished.

The contribution of different lymphocyte populations to glioma surveillance was examined through transplantation and in vivo immune-cell depletion experiments. Tumor-immune interactions were further characterized by transcriptomic profiling and flow cytometry in in vitro co-culture assays. Our findings identified CD4⁺ T lymphocytes as key initiators of the immune response against glioma cells, promoting the recruitment of cytotoxic CD8⁺ T cells. Nevertheless, during malignant progression, glioma cells acquired the ability to interfere with this immune response. Compared with their low-grade counterparts, high-grade glioma cells showed decreased expression of genes associated with MHC class II-mediated antigen presentation and reduced interferon signaling. Moreover, they directly impaired cytotoxic immune populations, decreasing the proportion of granzyme B-expressing CD8⁺ T lymphocytes and NK cells through mechanisms requiring cell-cell contact.

The immunomodulatory activity of high-grade glioma cells also affected neighboring tumor cells. Indeed, their co-transplantation promoted the engraftment of low-grade glioma cells that were otherwise immunogenic and unable to generate tumors.

Collectively, these results indicate that glioma progression does not depend exclusively on tumor-intrinsic mechanisms that allow malignant cells to remain undetected. Instead, high-grade glioma cells actively reshape the immune microenvironment, suppress cytotoxic responses, and establish conditions permissive for tumor growth.



Supported by SciSerTec

Web: <http://www.scisertec.de>